



NOTRE DAME UNIVERSITY

BANGLADESH

Machine Learning Lab Report-05

Course Code: CSE4214

Course Title: Machine Learning Lab

Lab Task Topic: Naive Bayes Theorem

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Submission Date: April 25, 2026

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1 Objective

The objective of this experiment is to implement the Naive Bayes classification algorithm using the `CategoricalNB` model from the Scikit-learn library. The model is used to predict whether a customer will make a purchase based on categorical features such as Day, Discount availability, and Free Delivery option. Additionally, the experiment aims to understand the process of data preprocessing, label encoding, model training, and evaluation of classification performance.

2 Dataset Description

The following datasets were used in this Naive Bayes lab experiment:

- **NaiveBayesDataset.csv**
This dataset is used for demonstrating the basic implementation of the Naive Bayes classification algorithm. It contains labeled features suitable for probability-based classification tasks.
- **titanic.csv**
This dataset contains passenger information from the Titanic disaster. It is commonly used for classification problems such as predicting survival based on features like age, gender, and passenger class.
- **cardio_train.csv**
This dataset includes medical data used to predict cardiovascular disease. It contains features such as age, blood pressure, cholesterol level, and lifestyle indicators.
- **feature_selection.csv**
This dataset is used to analyze feature importance and selection techniques. It helps in identifying the most relevant attributes for improving classification performance.

Since all features are categorical in nature, label encoding is applied to convert them into numerical form before training the Naive Bayes classifier.

3 Implementing Naive Bayes Theorem

3.1 Data Loading and Library Import for Naive Bayes

Explanation

In this step, we import essential Python libraries required for building a Naive Bayes classification model. `pandas` is used for data handling and analysis. `train_test_split` helps to divide the dataset into training and testing sets. `CategoricalNB` is the Naive Bayes classifier suitable for categorical features. `LabelEncoder` is used to convert categorical variables into numerical form. Finally, `accuracy_score` and `classification_report` are used to evaluate model performance.

After importing the libraries, the dataset `NaiveBayesDataset.csv` is loaded using `pd.read_csv()`, and displayed to verify its structure and contents.

```
[1]: import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import CategoricalNB
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, classification_report
```

```
[2]: # Load the dataset
df = pd.read_csv('NaiveBayesDataset.csv')
```

```
[3]: df
```

Output

The dataset is successfully loaded into a pandas DataFrame. The output displays the first view of the dataset in tabular form, showing rows and columns with categorical features and the target label. This confirms that the data is correctly imported and ready for preprocessing and model training.

```
[3]:
```

	Day	Discount	Free_Delivery	Purchase
0	Weekday	Yes	Yes	Yes
1	Weekday	Yes	Yes	Yes
2	Weekday	No	No	No
3	Holiday	Yes	Yes	Yes
4	Weekend	Yes	Yes	Yes
5	Holiday	No	No	No
6	Weekend	Yes	No	Yes
7	Weekday	Yes	Yes	Yes
8	Weekend	Yes	Yes	Yes
9	Holiday	Yes	Yes	Yes
10	Holiday	No	Yes	Yes
11	Holiday	No	No	No
12	Weekend	Yes	Yes	Yes
13	Holiday	Yes	Yes	Yes
14	Holiday	Yes	Yes	Yes
15	Weekday	Yes	Yes	Yes
16	Holiday	No	Yes	Yes
17	Weekday	Yes	No	Yes
18	Weekend	No	No	Yes
19	Weekend	No	Yes	Yes
20	Weekday	Yes	Yes	Yes
21	Weekend	Yes	Yes	No
22	Holiday	No	Yes	Yes
23	Weekday	Yes	Yes	Yes
24	Holiday	No	No	No
25	Weekday	No	Yes	No
26	Weekday	Yes	Yes	Yes
27	Weekday	Yes	Yes	Yes
28	Holiday	Yes	Yes	Yes
29	Weekend	Yes	Yes	Yes

```
[4]: # Encode categorical variables
label_encoders = {}
for column in ['Day', 'Discount', 'Free_Delivery', 'Purchase']:
    le = LabelEncoder()
    df[column] = le.fit_transform(df[column])
    label_encoders[column] = le
```

```
[5]: # Split data into features (X) and target (y)
X = df[['Day', 'Discount', 'Free_Delivery']]
y = df['Purchase']
```

```
[6]: # Split into training and test sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2,
↳ random_state=42)
```

```
[7]: # Initialize and train the Naive Bayes model
nb_model = CategoricalNB()
nb_model.fit(X_train, y_train)
```

```
[7]: CategoricalNB()
```

```
[8]: # Make predictions on the test set
y_pred = nb_model.predict(X_test)
```

```
[9]: # Evaluate the model
print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

Accuracy: 1.0

	precision	recall	f1-score	support
1	1.00	1.00	1.00	6
accuracy			1.00	6
macro avg	1.00	1.00	1.00	6
weighted avg	1.00	1.00	1.00	6

```
[10]: # Example prediction
example = pd.DataFrame({'Day': ['Holiday'], 'Discount': ['Yes'],
↳ 'Free_Delivery': ['No']})
for column in example.columns:
    example[column] = label_encoders[column].transform(example[column])
prediction = nb_model.predict(example)
print("Purchase Prediction:", label_encoders['Purchase'].
↳ inverse_transform(prediction))
```

Purchase Prediction: ['Yes']

```
[11]: # Check if "Holiday" exists in the Day encoder classes
print("Classes for Day:", label_encoders['Day'].classes_)
```

Classes for Day: ['Holiday' 'Weekday' 'Weekend']

```
[12]: # Example prediction
example = pd.DataFrame({'Day': ['Holiday'], 'Discount': ['Yes'],
↳ 'Free_Delivery': ['No']})
# This line creates a DataFrame named example with a single row representing
↳ an example input.
```

```
[13]: # Encode the example input using updated encoders
for column in example.columns:
    example[column] = label_encoders[column].transform(example[column])
#Purpose: This loop iterates over each column in the example DataFrame and
↳ transforms the categorical
#text values into numeric codes, which the model requires as input.
```

```
[14]: # Make prediction
prediction = nb_model.predict(example)
#Purpose: This line uses the Naive Bayes model (nb_model) to make a
↳ prediction
# on the transformed example input.
```

```
[15]: # Decode the prediction back to original label
print("Purchase Prediction:", label_encoders['Purchase'].
↳ inverse_transform(prediction))
```

Purchase Prediction: ['Yes']

```
[16]: # Example prediction
example = pd.DataFrame({'Day': ['Weekend'], 'Discount': ['Yes'],
↳ 'Free_Delivery': ['Yes']})
# This line creates a DataFrame named example with a single row representing
↳ an example input
```

```
[17]: # Encode the example input using updated encoders
for column in example.columns:
    example[column] = label_encoders[column].transform(example[column])
#Purpose: This loop iterates over each column in the example DataFrame and
↳ transforms the categorical
#text values into numeric codes, which the model requires as input.
```

```
[18]: # Make prediction
prediction = nb_model.predict(example)
#Purpose: This line uses the Naive Bayes model (nb_model) to make a
↳ prediction
# on the transformed example input.
```

```
[19]: # Decode the prediction back to original label
print("Purchase Prediction:", label_encoders['Purchase'].
↳ inverse_transform(prediction))
```

Purchase Prediction: ['Yes']

4 Naive Bayes Classifier - Titanic

```
[20]: import pandas as pd
```

```
[21]: df= pd.read_csv('titanic.csv')
df.head()
```

```
[21]: PassengerId  Survived  Pclass  \
0          892         0         3
1          893         1         3
2          894         0         2
3          895         0         3
4          896         1         3
```

```

                                Name      Sex  Age  SibSp  Parch  \
↪\
0                                Kelly, Mr. James    male  34.5      0      0
1                Wilkes, Mrs. James (Ellen Needs)  female  47.0      1      0
2                  Myles, Mr. Thomas Francis    male  62.0      0      0
3                                Wirz, Mr. Albert    male  27.0      0      0
4  Hirvonen, Mrs. Alexander (Helga E Lindqvist)  female  22.0      1      1
```

```

Ticket      Fare  Cabin  Embarked
0  330911    7.8292   NaN        Q
1  363272    7.0000   NaN        S
2  240276    9.6875   NaN        Q
3  315154    8.6625   NaN        S
4  3101298  12.2875   NaN        S
```

```
[22]: df.isnull().sum()
```

```
[22]: PassengerId      0
Survived             0
Pclass              0
Name                0
Sex                0
Age                86
SibSp              0
Parch              0
Ticket             0
Fare               1
Cabin             327
Embarked           0
dtype: int64
```

```
[23]: # removing null values of Age// better if we remove later on the updated
↪dataset
handle = df['Age'].mean()
handle
```

```
[23]: np.float64(30.272590361445783)
```

```
[24]: df.Age= df.Age.fillna(handle)
```

```
[25]: df
```

```
[25]: PassengerId  Survived  Pclass  \
0          892         0         3
```

1	893	1	3
2	894	0	2
3	895	0	3
4	896	1	3
..	...	...	...
413	1305	0	3
414	1306	1	1
415	1307	0	3
416	1308	0	3
417	1309	0	3

	Name	Sex	Age	SibSp	\
0	Kelly, Mr. James	male	34.50000	0	
1	Wilkes, Mrs. James (Ellen Needs)	female	47.00000	1	
2	Myles, Mr. Thomas Francis	male	62.00000	0	
3	Wirz, Mr. Albert	male	27.00000	0	
4	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.00000	1	
..	...	...	...	...	
413	Spector, Mr. Woolf	male	30.27259	0	
414	Oliva y Ocana, Dona. Fermina	female	39.00000	0	
415	Saether, Mr. Simon Sivertsen	male	38.50000	0	
416	Ware, Mr. Frederick	male	30.27259	0	
417	Peter, Master. Michael J	male	30.27259	1	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	330911	7.8292	NaN	Q
1	0	363272	7.0000	NaN	S
2	0	240276	9.6875	NaN	Q
3	0	315154	8.6625	NaN	S
4	1	3101298	12.2875	NaN	S
..	...	...	...	...	...
413	0	A.5. 3236	8.0500	NaN	S
414	0	PC 17758	108.9000	C105	C
415	0	SOTON/O.Q. 3101262	7.2500	NaN	S
416	0	359309	8.0500	NaN	S
417	1	2668	22.3583	NaN	C

[418 rows x 12 columns]

```
[26]: # Assumption: Make a naive assumption that features such as
      ↪ male, class, age, cabin, fare, etc. are independent of each other
```

```
[27]: df.
      ↪ drop(['PassengerId', 'Name', 'SibSp', 'Parch', 'Ticket', 'Cabin', 'Embarked'], axis=
      ↪ 'columns', inplace= True)
```

```
[28]: df.head()
```

```
[28]:
```

	Survived	Pclass	Sex	Age	Fare
0	0	3	male	34.5	7.8292
1	1	3	female	47.0	7.0000

2	0	2	male	62.0	9.6875
3	0	3	male	27.0	8.6625
4	1	3	female	22.0	12.2875

```
[29]: # 1 - survived 0 - not survived , At the end we will work on final
      # obtained from inputs and targer for train_test_split()
```

```
[30]: # 1 - survived 0 - not survived
      inputs= df.drop('Survived', axis = 'columns')
      target = df.Survived
```

```
[31]: # Encoding Sex
      dummies = pd.get_dummies(df['Sex'])
```

```
[32]: dummies = dummies.astype(int)
      # will display 0 and 1 instead of True and False
```

```
[33]: dummies
```

```
[33]:      female  male
0         0     1
1         1     0
2         0     1
3         0     1
4         1     0
..      ...  ...
413        0     1
414        1     0
415        0     1
416        0     1
417        0     1

[418 rows x 2 columns]
```

```
[34]: #inputs = inputs.drop('Sex','male' axis='columns')
```

```
[35]: df
```

```
[35]:      Survived  Pclass    Sex      Age      Fare
0           0        3   male  34.50000    7.8292
1           1        3  female  47.00000    7.0000
2           0        2   male  62.00000    9.6875
3           0        3   male  27.00000    8.6625
4           1        3  female  22.00000   12.2875
..      ...      ...      ...      ...      ...
413         0        3   male  30.27259    8.0500
414         1        1  female  39.00000  108.9000
415         0        3   male  38.50000    7.2500
416         0        3   male  30.27259    8.0500
417         0        3   male  30.27259   22.3583
```

```
[418 rows x 5 columns]
```

```
[36]: inputs
```

```
[36]:
```

	Pclass	Sex	Age	Fare
0	3	male	34.50000	7.8292
1	3	female	47.00000	7.0000
2	2	male	62.00000	9.6875
3	3	male	27.00000	8.6625
4	3	female	22.00000	12.2875
..	...	...	...	...
413	3	male	30.27259	8.0500
414	1	female	39.00000	108.9000
415	3	male	38.50000	7.2500
416	3	male	30.27259	8.0500
417	3	male	30.27259	22.3583

```
[418 rows x 4 columns]
```

```
[37]: inputs = inputs.drop(['Sex'],axis='columns')
```

```
[38]: inputs
```

```
[38]:
```

	Pclass	Age	Fare
0	3	34.50000	7.8292
1	3	47.00000	7.0000
2	2	62.00000	9.6875
3	3	27.00000	8.6625
4	3	22.00000	12.2875
..	...	...	...
413	3	30.27259	8.0500
414	1	39.00000	108.9000
415	3	38.50000	7.2500
416	3	30.27259	8.0500
417	3	30.27259	22.3583

```
[418 rows x 3 columns]
```

```
[39]: #concatening columns with sex column
df
```

```
[39]:
```

	Survived	Pclass	Sex	Age	Fare
0	0	3	male	34.50000	7.8292
1	1	3	female	47.00000	7.0000
2	0	2	male	62.00000	9.6875
3	0	3	male	27.00000	8.6625
4	1	3	female	22.00000	12.2875
..	...	...	...	...	...
413	0	3	male	30.27259	8.0500
414	1	1	female	39.00000	108.9000
415	0	3	male	38.50000	7.2500

```

416      0      3   male  30.27259    8.0500
417      0      3   male  30.27259   22.3583

```

```
[418 rows x 5 columns]
```

```
[40]: inputs
```

```

[40]:      Pclass      Age      Fare
0         3  34.50000    7.8292
1         3  47.00000    7.0000
2         2  62.00000    9.6875
3         3  27.00000    8.6625
4         3  22.00000   12.2875
..      ...      ...      ...
413        3  30.27259    8.0500
414        1  39.00000  108.9000
415        3  38.50000    7.2500
416        3  30.27259    8.0500
417        3  30.27259   22.3583

```

```
[418 rows x 3 columns]
```

```
[41]: merged = pd.concat([inputs,dummies],axis = 'columns')
```

```
[42]: merged
```

```

[42]:      Pclass      Age      Fare  female  male
0         3  34.50000    7.8292         0     1
1         3  47.00000    7.0000         1     0
2         2  62.00000    9.6875         0     1
3         3  27.00000    8.6625         0     1
4         3  22.00000   12.2875         1     0
..      ...      ...      ...      ...
413        3  30.27259    8.0500         0     1
414        1  39.00000  108.9000         1     0
415        3  38.50000    7.2500         0     1
416        3  30.27259    8.0500         0     1
417        3  30.27259   22.3583         0     1

```

```
[418 rows x 5 columns]
```

```
[43]: final = merged.drop(['male'],axis = 'columns')
```

```
[44]: final
```

```

[44]:      Pclass      Age      Fare  female
0         3  34.50000    7.8292         0
1         3  47.00000    7.0000         1
2         2  62.00000    9.6875         0
3         3  27.00000    8.6625         0
4         3  22.00000   12.2875         1

```

```

..      ...      ...      ...      ...
413      3  30.27259      8.0500      0
414      1  39.00000     108.9000      1
415      3  38.50000      7.2500      0
416      3  30.27259      8.0500      0
417      3  30.27259     22.3583      0

```

```
[418 rows x 4 columns]
```

```
[45]: final.columns[final.isna().any()]
```

```
[45]: Index(['Fare'], dtype='str')
```

```
[46]: null_count_city = df['Fare'].isnull().sum()
print(f"Number of null values in 'Fare' column: {null_count_city}")
```

```
Number of null values in 'Fare' column: 1
```

```
[47]: final.Fare[:50]
```

```
[47]: 0      7.8292
1      7.0000
2      9.6875
3      8.6625
4     12.2875
5      9.2250
6      7.6292
7     29.0000
8      7.2292
9     24.1500
10     7.8958
11     26.0000
12     82.2667
13     26.0000
14     61.1750
15     27.7208
16     12.3500
17      7.2250
18      7.9250
19      7.2250
20     59.4000
21      3.1708
22     31.6833
23     61.3792
24    262.3750
25     14.5000
26     61.9792
27      7.2250
28     30.5000
29     21.6792
30     26.0000
31     31.5000
```

```

32      20.5750
33      23.4500
34      57.7500
35       7.2292
36       8.0500
37       8.6625
38       9.5000
39      56.4958
40      13.4167
41      26.5500
42       7.8500
43      13.0000
44      52.5542
45       7.9250
46      29.7000
47       7.7500
48      76.2917
49      15.9000

```

Name: Fare, dtype: float64

```
[48]: final.Fare = final.Fare.fillna(final.Fare.mean())
      final.head()
```

```
[48]:
```

	Pclass	Age	Fare	female
0	3	34.5	7.8292	0
1	3	47.0	7.0000	1
2	2	62.0	9.6875	0
3	3	27.0	8.6625	0
4	3	22.0	12.2875	1

```
[49]: from sklearn.model_selection import train_test_split
```

```
[50]: xtrain, xtest, ytrain, ytest = train_test_split(final, target, test_size = 0.
      ↪30, random_state =1)
```

```
[51]: from sklearn.naive_bayes import GaussianNB
      model = GaussianNB ()
```

```
[52]: model.fit(xtrain, ytrain)
```

```
[52]: GaussianNB()
```

```
[53]: model.score(xtest,ytest)
```

```
[53]: 1.0
```

```
[54]: xtest[:10]
```

```
[54]:
```

	Pclass	Age	Fare	female
358	3	30.27259	7.7500	0
164	2	41.00000	13.0000	0

17	3	21.00000	7.2250	0
67	1	47.00000	42.4000	0
4	3	22.00000	12.2875	1
377	2	21.00000	11.5000	0
214	3	38.00000	7.7750	1
290	1	30.27259	39.6000	0
381	3	26.00000	7.8792	0
5	3	14.00000	9.2250	0

```
[55]: ytest[0:10]
```

```
[55]: 358    0
      164    0
      17    0
      67    0
       4    1
      377    0
      214    1
      290    0
      381    0
       5    0
      Name: Survived, dtype: int64
```

```
[56]: model.predict(xtest[0:10])
```

```
[56]: array([0, 0, 0, 0, 1, 0, 1, 0, 0, 0])
```

```
[57]: model.predict_proba(xtest[:10])
```

```
[57]: array([[1., 0.],
            [1., 0.],
            [1., 0.],
            [1., 0.],
            [0., 1.],
            [1., 0.],
            [0., 1.],
            [1., 0.],
            [1., 0.],
            [1., 0.]])
```

```
[58]: #pclass, Age, Fare, Gender
```

```
test = [[1,23.000000,113.2750,0]]
#test = xtest
a= model.predict(test)
if a[0] == 0:
    print("Not Survived")
else:
    print(" Survived")
```

Not Survived

```

/media/spyder/Secondary-Disk/Github/Student-Performance-Prediction-with-
Explainable-AI/CodeBase/Trial/.spenv/lib/python3.11/site-
packages/sklearn/base.py:493: UserWarning: X does not have valid feature_
↪names,
but GaussianNB was fitted with feature names
warnings.warn(

```

4.1 Calculate value with cross validation

```
[59]: from sklearn.model_selection import cross_val_score
```

```
[60]: cross_val_score (GaussianNB(),xtrain,ytrain,cv = 5)
```

```
[60]: array([1., 1., 1., 1., 1.])
```

5 Feature-selection-stdm

5.1 Univariate Selecection Method @Feature Selection, Algorithm: SelectKBest Algo- rithm

```
[61]: import pandas as pd
import numpy as np
import seaborn as sns
```

```
[62]: #df = pd.read_csv("bank.csv",sep = ';')
df = pd.read_csv("cardio_train.csv", sep = ';')
```

```
[63]: df
```

```
[63]:
```

	id	age	gender	height	weight	ap_hi	ap_lo	cholesterol	gluc
0	0	18393	2	168	62.0	110	80	1	1
1	1	20228	1	156	85.0	140	90	3	1
2	2	18857	1	165	64.0	130	70	3	1
3	3	17623	2	169	82.0	150	100	1	1
4	4	17474	1	156	56.0	100	60	1	1
...	...	...	...	...	...	...	...	...	...
69995	99993	19240	2	168	76.0	120	80	1	1
69996	99995	22601	1	158	126.0	140	90	2	2
69997	99996	19066	2	183	105.0	180	90	3	1
69998	99998	22431	1	163	72.0	135	80	1	2
69999	99999	20540	1	170	72.0	120	80	2	1

	smoke	alco	active	cardio
0	0	0	1	0
1	0	0	1	1
2	0	0	0	1
3	0	0	1	1
4	0	0	0	0
...	...	...	...	...

69995	1	0	1	0
69996	0	0	1	1
69997	0	1	0	1
69998	0	0	0	1
69999	0	0	1	0

[70000 rows x 13 columns]

```
[64]: df.shape
```

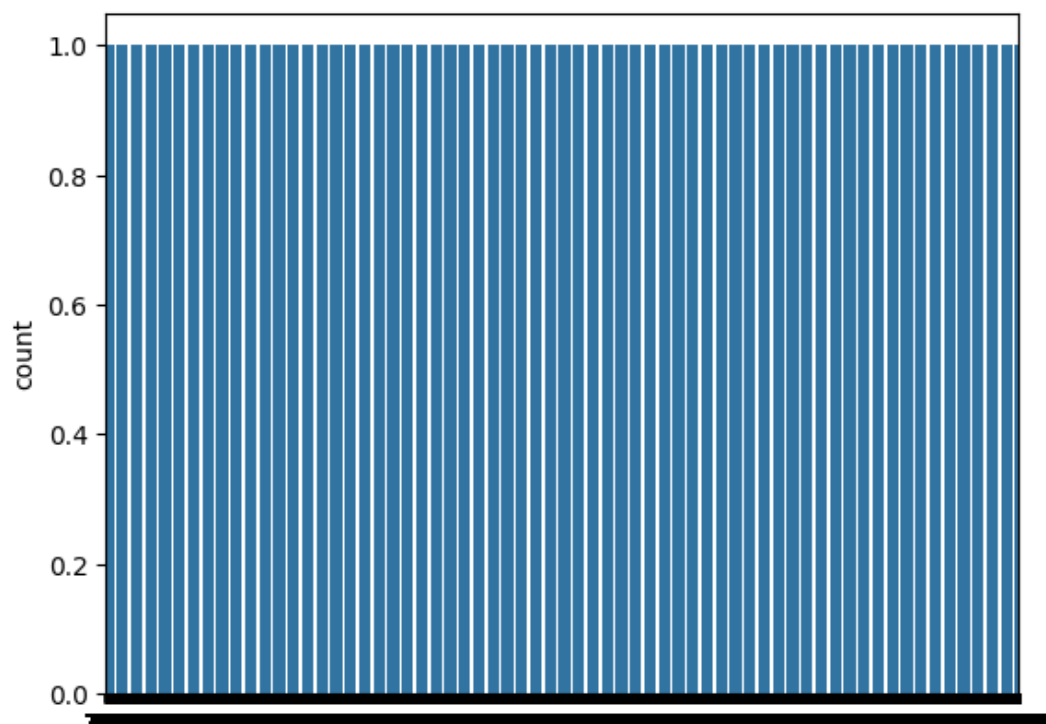
```
[64]: (70000, 13)
```

```
[65]: df["cardio"].value_counts()
```

```
[65]: cardio  
0    35021  
1    34979  
Name: count, dtype: int64
```

```
[66]: sns.countplot(df["cardio"])
```

```
[66]: <Axes: ylabel='count'>
```



- `x = df.iloc[:, :-1]:`
- This line is selecting all rows (:) and all columns except the last one (:-1).
- `:` means “select all rows.”

- `:-1` means “select all columns except the last one.” In Python slicing, `:-1` excludes the last item, so this will include all columns up to, but not including, the last one.
- Result: `x` will be a DataFrame containing all the columns of `df` except the last column

```
[67]: x = df.iloc[:, :-1]
```

- `y = df.iloc[:, 12]`
- This line is selecting all rows (`:`) and the 13th column (remember, indexing in Python starts at 0, so index 12 corresponds to the 13th column).
- `:` means “select all rows.”
- 12 means “select the 13th column” (as Python uses 0-based indexing).
- Result: `y` will be a Series containing the values of the 13th column of `df`.

```
[68]: # y = df.iloc[:, 12]
y = df.iloc[:, 12]
```

```
[69]: from sklearn.feature_selection import SelectKBest
```

```
[70]: from sklearn.feature_selection import f_classif
```

```
[71]: FIT_FEATURES = SelectKBest(score_func = f_classif)
```

```
[72]: FIT_FEATURES.fit(x,y)
```

```
[72]: SelectKBest()
```

```
[73]: pd.DataFrame(FIT_FEATURES.scores_)
```

```
[73]:
          0
0      1.010461
1    4209.007957
2      4.603641
3      8.197397
4    2388.777887
5      208.339524
6     303.629011
7    3599.361137
8      562.772977
9      16.790541
10     3.761355
11     89.091494
```

```
[74]: score_COL = pd.DataFrame(FIT_FEATURES.scores_)
```

```
[75]: score_COL
```

```
[75]:
          0
0      1.010461
1    4209.007957
2      4.603641
3      8.197397
4    2388.777887
```

```
5    208.339524
6    303.629011
7    3599.361137
8    562.772977
9     16.790541
10     3.761355
11    89.091494
```

```
[76]: # Assuming FIT_FEATURES_scores is an array or a list of scores
score_COL = pd.DataFrame(FIT_FEATURES.scores_, columns=["scorevalue"])
# Now score_COL is a DataFrame with a single column named "scorevalue"
```

```
[77]: score_COL
```

```
[77]:    scorevalue
0      1.010461
1  4209.007957
2      4.603641
3      8.197397
4  2388.777887
5    208.339524
6    303.629011
7  3599.361137
8    562.772977
9     16.790541
10     3.761355
11    89.091494
```

```
[78]: NEW_COL = pd.DataFrame(x.columns)
```

```
[79]: NEW_COL
```

```
[79]:      0
0      id
1      age
2      gender
3      height
4      weight
5      ap_hi
6      ap_lo
7  cholesterol
8      gluc
9      smoke
10     alco
11     active
```

```
[80]: top_features = pd.concat([NEW_COL, score_COL], axis=1)
```

```
[81]: top_features
```

```
[81]:
      0    scorevalue
0      id    1.010461
1     age  4209.007957
2   gender    4.603641
3   height    8.197397
4   weight  2388.777887
5    ap_hi   208.339524
6    ap_lo   303.629011
7 cholesterol  3599.361137
8     gluc   562.772977
9    smoke   16.790541
10    alco    3.761355
11   active   89.091494
```

```
[82]: top_features.nlargest(8, "scorevalue")
```

```
[82]:
      0    scorevalue
1     age  4209.007957
7 cholesterol  3599.361137
4   weight  2388.777887
8     gluc   562.772977
6    ap_lo   303.629011
5    ap_hi   208.339524
11   active   89.091494
9    smoke   16.790541
```

6 Feature-selection-using-correlation

```
[83]: import pandas as pd
      # Load the dataset
      df = pd.read_csv('feature_selection.csv')
```

```
[84]: df
```

```
[84]: location    area  bedroom  bathroom owner  blodd group  price
0         a    dhaka         3         2      x      o+  10000
1         b    dhaka         4         2      y      ab+  12000
2         c  khulna         2         2      p      ab-  11000
3         d rajshahi         3         2      q      o+  15000
4         e  cumilla         3         1      w      ab+  12000
5         f    dhaka         2         1      l      ab-  11000
6         g barishal         2         1      r      o+  10000
7         h  natore         2         3      s      o+  15000
8         i  khulna         3         3      k      ab+  12000
9         j    dhaka         4         2      v      ab-  11000
```

```
[85]: # Check for non-numeric data in the columns
      print(df.dtypes)
```

```
location    str
area        str
```

```

bedroom      int64
bathroom     int64
owner        str
blodd group  str
price        int64
dtype: object

```

```

[86]: # Identify non-numeric values in specific columns
      for column in df.columns:
          if df[column].dtype == 'object':
              print(f"Non-numeric values in '{column}':\n", df[column].unique())

```

```

[87]: # Option 1: Convert columns to numeric, forcing errors to NaN
      df = df.apply(pd.to_numeric, errors='coerce')

```

```

[88]: df

```

```

[88]:   location  area  bedroom  bathroom  owner  blodd group  price
0      NaN    NaN      3         2    NaN      NaN    10000
1      NaN    NaN      4         2    NaN      NaN    12000
2      NaN    NaN      2         2    NaN      NaN    11000
3      NaN    NaN      3         2    NaN      NaN    15000
4      NaN    NaN      3         1    NaN      NaN    12000
5      NaN    NaN      2         1    NaN      NaN    11000
6      NaN    NaN      2         1    NaN      NaN    10000
7      NaN    NaN      2         3    NaN      NaN    15000
8      NaN    NaN      3         3    NaN      NaN    12000
9      NaN    NaN      4         2    NaN      NaN    11000

```

```

[89]: # Calculate correlation with the target variable 'price'
      correlation = df.corr()

```

```

[90]: print(df.dtypes)

```

```

location      float64
area          float64
bedroom       int64
bathroom      int64
owner         float64
blodd group   float64
price         int64
dtype: object

```

```

[91]: # Option 1: Convert columns to numeric, forcing errors to NaN
      df = df.apply(pd.to_numeric, errors='coerce')

```

```

[92]: # Calculate correlation with the target variable 'price'
      correlation = df.corr()

```

```

[93]: # Display the correlation of each feature with the target variable 'price'
      correlated_features = correlation['price'].sort_values(ascending=False)
      print("Correlation with Price:\n", correlated_features)

```

```
Correlation with Price:
price          1.000000
bathroom       0.495798
bedroom        -0.015721
location       NaN
area           NaN
owner          NaN
blodd group    NaN
Name: price, dtype: float64
```

6.1 Encoding / Handling another column => location

```
[94]: from sklearn.preprocessing import OrdinalEncoder
      # Load the dataset
      df = pd.read_csv('feature_selection.csv')
```

```
[95]: categorical_column = 'location'
```

```
[96]: df
```

```
[96]:
```

	location	area	bedroom	bathroom	owner	blodd group	price
0	a	dhaka	3	2	x	o+	10000
1	b	dhaka	4	2	y	ab+	12000
2	c	khulna	2	2	p	ab-	11000
3	d	rajshahi	3	2	q	o+	15000
4	e	cumilla	3	1	w	ab+	12000
5	f	dhaka	2	1	l	ab-	11000
6	g	barishal	2	1	r	o+	10000
7	h	natore	2	3	s	o+	15000
8	i	khulna	3	3	k	ab+	12000
9	j	dhaka	4	2	v	ab-	11000

```
[97]: print(df.columns)
```

```
Index(['location ', 'area ', 'bedroom ', 'bathroom', 'owner ', 'blodd group',
      'price'],
      dtype='str')
```

```
[98]: df.columns = df.columns.str.strip() # Remove any leading/trailing spaces
      df.columns = df.columns.str.lower() # Convert all column names to lowercase
      ↪(optional)
      print(df.columns)
```

```
Index(['location', 'area', 'bedroom', 'bathroom', 'owner', 'blodd group',
      'price'],
      dtype='str')
```

```
[99]: # Optional: Use a conditional check
      if 'location' in df.columns:
          print(df['location'].head())
      else:
          print("The 'location' column is not in the DataFrame.")
```

```

0    a
1    b
2    c
3    d
4    e
Name: location, dtype: str

```

```
[100]: encoder = OrdinalEncoder()
df[categorical_column] = encoder.fit_transform(df[[categorical_column]])
```

```
[101]: df
```

```
[101]:
```

	location	area	bedroom	bathroom	owner	blodd	group	price
0	0.0	dhaka	3	2	x		o+	10000
1	1.0	dhaka	4	2	y		ab+	12000
2	2.0	khulna	2	2	p		ab-	11000
3	3.0	rajshahi	3	2	q		o+	15000
4	4.0	cumilla	3	1	w		ab+	12000
5	5.0	dhaka	2	1	l		ab-	11000
6	6.0	barishal	2	1	r		o+	10000
7	7.0	natore	2	3	s		o+	15000
8	8.0	khulna	3	3	k		ab+	12000
9	9.0	dhaka	4	2	v		ab-	11000

7 Feature Selection using Correlation -U

```
[102]: import pandas as pd
# Load the dataset
df = pd.read_csv('feature_selection.csv')
```

```
[103]: df
```

```
[103]:
```

	location	area	bedroom	bathroom	owner	blodd	group	price
0	a	dhaka	3	2	x		o+	10000
1	b	dhaka	4	2	y		ab+	12000
2	c	khulna	2	2	p		ab-	11000
3	d	rajshahi	3	2	q		o+	15000
4	e	cumilla	3	1	w		ab+	12000
5	f	dhaka	2	1	l		ab-	11000
6	g	barishal	2	1	r		o+	10000
7	h	natore	2	3	s		o+	15000
8	i	khulna	3	3	k		ab+	12000
9	j	dhaka	4	2	v		ab-	11000

```
[104]: # Check for non-numeric data in the columns
print(df.dtypes)
```

```

location      str
area          str
bedroom      int64
bathroom      int64

```

```
owner          str
blodd group    str
price          int64
dtype: object
```

```
[105]: # Identify non-numeric values in specific columns
for column in df.columns:
    if df[column].dtype == 'object':
        print(f"Non-numeric values in '{column}':\n", df[column].unique())
```

7.1 Encoding / Handling another column => location

```
[106]: from sklearn.preprocessing import OrdinalEncoder
categorical_column = 'location'
df
```

```
[106]:
```

	location	area	bedroom	bathroom	owner	blodd group	price
0	a	dhaka	3	2	x	o+	10000
1	b	dhaka	4	2	y	ab+	12000
2	c	khulna	2	2	p	ab-	11000
3	d	rajshahi	3	2	q	o+	15000
4	e	cumilla	3	1	w	ab+	12000
5	f	dhaka	2	1	l	ab-	11000
6	g	barishal	2	1	r	o+	10000
7	h	natore	2	3	s	o+	15000
8	i	khulna	3	3	k	ab+	12000
9	j	dhaka	4	2	v	ab-	11000

```
[107]: print(df.columns)

Index(['location ', 'area ', 'bedroom ', 'bathroom', 'owner ', 'blodd group',
      'price'],
      dtype='str')
```

```
[108]: df.columns = df.columns.str.strip() # Remove any leading/trailing spaces
df.columns = df.columns.str.lower() # Convert all column names to lowercase
print(df.columns)

Index(['location', 'area', 'bedroom', 'bathroom', 'owner', 'blodd group',
      'price'],
      dtype='str')
```

```
[109]: # Optional: Use a conditional check
if 'location' in df.columns:
    print(df['location'].head())
else:
    print("The 'location' column is not in the DataFrame.")
```

```
0    a
1    b
2    c
3    d
```

```
4     e
Name: location, dtype: str
```

```
[110]: encoder = OrdinalEncoder()
df[categorical_column] = encoder.fit_transform(df[[categorical_column]])
```

```
[111]: df
```

```
[111]:
```

	location	area	bedroom	bathroom	owner	blodd	group	price
0	0.0	dhaka	3	2	x		o+	10000
1	1.0	dhaka	4	2	y		ab+	12000
2	2.0	khulna	2	2	p		ab-	11000
3	3.0	rajshahi	3	2	q		o+	15000
4	4.0	cumilla	3	1	w		ab+	12000
5	5.0	dhaka	2	1	l		ab-	11000
6	6.0	barishal	2	1	r		o+	10000
7	7.0	natore	2	3	s		o+	15000
8	8.0	khulna	3	3	k		ab+	12000
9	9.0	dhaka	4	2	v		ab-	11000

7.2 Encoding / Handling another column => area

```
[112]: categorical_column = 'area'
```

```
[113]: encoder = OrdinalEncoder()
df[categorical_column] = encoder.fit_transform(df[[categorical_column]])
```

```
[114]: df
```

```
[114]:
```

	location	area	bedroom	bathroom	owner	blodd	group	price
0	0.0	2.0	3	2	x		o+	10000
1	1.0	2.0	4	2	y		ab+	12000
2	2.0	3.0	2	2	p		ab-	11000
3	3.0	5.0	3	2	q		o+	15000
4	4.0	1.0	3	1	w		ab+	12000
5	5.0	2.0	2	1	l		ab-	11000
6	6.0	0.0	2	1	r		o+	10000
7	7.0	4.0	2	3	s		o+	15000
8	8.0	3.0	3	3	k		ab+	12000
9	9.0	2.0	4	2	v		ab-	11000

7.3 Encoding more than one column at a time

```
[115]: categorical_columns = ['owner', 'blodd group']
```

```
[116]: for column in categorical_columns:
        if column not in df.columns:
            print(f"Column '{column}' does not exist in the DataFrame")
        else:
            print(f"Encoding column: {column}")
```


Encoding column: owner

Encoding column: blodd group

```
[117]: encoder = OrdinalEncoder()
existing_categorical_columns = [col for col in categorical_columns if col in
    ↪df]
df[existing_categorical_columns] = encoder.
    ↪fit_transform(df[existing_categorical_columns])
```

```
[118]: df
```

```
[118]:
```

	location	area	bedroom	bathroom	owner	blodd group	price
0	0.0	2.0	3	2	8.0	2.0	10000
1	1.0	2.0	4	2	9.0	0.0	12000
2	2.0	3.0	2	2	2.0	1.0	11000
3	3.0	5.0	3	2	3.0	2.0	15000
4	4.0	1.0	3	1	7.0	0.0	12000
5	5.0	2.0	2	1	1.0	1.0	11000
6	6.0	0.0	2	1	4.0	2.0	10000
7	7.0	4.0	2	3	5.0	2.0	15000
8	8.0	3.0	3	3	0.0	0.0	12000
9	9.0	2.0	4	2	6.0	1.0	11000

```
[119]: # Calculate correlation with the target variable 'price'
correlation = df.corr()
```

```
[120]: # Display the correlation of each feature with the target variable 'price'
correlated_features = correlation['price'].sort_values(ascending=False)
print("Correlation with Price:\n", correlated_features)
```

```
Correlation with Price:
price          1.000000
area           0.797921
bathroom       0.495798
blodd group    0.148712
location       0.133118
bedroom        -0.015721
owner          -0.092159
Name: price, dtype: float64
```